Year 12 Maths - Applications of differentiation

Similar questions - Optimisation problems

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A square piece of card of side 30 cm has a square of side x cm cut from each corner. The remaining card is folded up to make an open box (a cuboid with no lid) of volume V cm^3.

(a) Show that $V = 900x - 120x^2 + 4x^3$. **[3]**

(b) Find the value of x that gives the maximum volume, and justify that it is a maximum. **[5]**

(c) Find the maximum volume. **[1]**

2 An open-topped rectangular tank has a square base of side x m and a volume of 32 m^3. The total area of the base and four sides is A m^2.

(a) Show that $A = x^2 + \dfrac{128}{x}$. **[3]**

(b) Find the minimum value of A, justifying that it is a minimum. **[5]**

(c) Find the height of the tank when A is a minimum. **[1]**

Question 1

(a) The base of the box has side $30-2x$ and the height is x. $V = x(30-2x)^2 = x(900-120x+4x^2) = 900x-120x^2+4x^3$

(b) $\dfrac{\mathrm{d}V}{\mathrm{d}x} = 900-240x+12x^2 = 12(x^2-20x+75) = 12(x-5)(x-15)$. $x = 15$ is impossible (the card would have no base), so $x = 5$. $\dfrac{\mathrm{d}^2V}{\mathrm{d}x^2} = -240+24x = -120 < 0$, so a maximum

(c) $V = 5 \times 20^2 = 2000$ cm^3

Question 2

(a) Height $h = \dfrac{32}{x^2}$. $A = x^2+4xh = x^2+\dfrac{128}{x}$

(b) $\dfrac{\mathrm{d}A}{\mathrm{d}x} = 2x-\dfrac{128}{x^2} = 0$ gives $x^3 = 64$, $x = 4$. $\dfrac{\mathrm{d}^2A}{\mathrm{d}x^2} = 2+\dfrac{256}{x^3} = 6 > 0$, so a minimum. $A = 16+32 = 48$ m^2

(c) $h = \dfrac{32}{16} = 2$ m